EAB 770 - Chapter 4

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1 \$ 2 $\times$r\$ and $s \times 2$ tables

1.1 Outline

- $2 \times r$ table
- Sets of $2 \times r$ tables
- $s \times 2$ table
- Sets of $s \times 2$ tables
  - Relationships between Sets of Tables
  - Exact Analysis of Association ($s \times 2$) tables)

2 what if there are more than two outcomes?

2.1 Multiple Categorical Outcomes

- Likert Scale (Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree)
- Part of a whole (None, Some, All)
- We would need more than two columns in our contingency table to account for these other outcomes.
  - Variables can be forced to be "binary", but some important information can be lost.
- If there are r outcome levels and 2 levels of the exploratory variable, we use a $2 \times r$ contingency table.
  - Focus on outcomes being ordinally scaled.

2.2 $2 \times r$ table

Let A_j be the j^{th} outcome level.

- Since A_J is ordinal, then $A_1 < A_2 < A_3 < ...A_r$.
  - The $\leq$ can replace $<$ as long as at least one is a strict inequality

To account for ordinality, we assign a set of scores **a** reflecting the relationship between response levels.

$$\mathbf{a} = \{a_1, a_2, a_3, ...a_r\} : a_1 < a_2 < ... < a_r$$

	A_1	A_2	...	A_r	Total
X_1	n_{11}	n_{12}	...	n_{1r}	n_{1+}
X_2	n_{21}	n_{22}	...	n_{2r}	n_{2+}
Total	n_{+1}	n_{+2}	...	n_{+r}	n

2.3 Row Mean Scores

We can define a group mean for X_1, denoted by $\bar{f_1}$.

$$\bar{f_1} = \sum_{j=1}^{r} \frac{a_j n_{1j}}{n_{1+}}$$

The random variable $\bar{f_1}$ approximately has a normal distribution when the central limit theorem is imposed.

- High row sum

We refer to these scores as **Row Mean Scores**

2.4 2 $\times$r table: Null hypothesis

Null hypothesis H_0: There is no association between the row (X) and column (A) variables.

Under the null hypothesis:

- The expected value of the mean can be calculated as:

$$\mu_a = \sum_{j=1}^{r} \frac{a_j n_{+j}}{n}$$

2.5 Mean Score Statistic

We can define a mean score statistic, Q_S, given by

$$Q_S = \frac{(\bar{f}_1 - \mu_1)^2}{V}$$

> **Note**
>
> - V is the variance of $\bar{f}_1$ under the null hypothesis.
> - Q_S follows a chi-square distribution with one degree of freedom.

2.6 Mean Score Statistic

- Targets alternative hypothesis of mean shifts to the hypothesis of no association
  - Uses fewer degrees of freedom compared to Pearson chi-square

> **Note**
>
> Pearson chi-square requires $(r-1)$ degrees of freedom compared to 1 degree of freedom for Q_S

- The mean score statistic can also be used to determine the tendency of one group having better scores compared to the other.
- The Q_S sample size condition is less stringent:

> **Note**
>
> Choose one or more cutpoints $ j = (2,3,...(r-1))$, add the first through j^{th} columns together and add the $(j-1)^{th}$ through r^{th} columns together. If both of these sums are 5 or greater for at least one cutpoint in each row, then the sample size is adequate.

2.7 $2 \times r$ analysis: SAS

- Data should be encoded in increasing order of the outcome variable.
- The `FREQ` procedure will give us the mean score statistic:
  - **ORDER=DATA should be included in the code with the `PROC FREQ` statement. (See sample code)**
- Can be obtained using the `cmh` option or `chisq` option in the `tables` statement
  - Q_S: Mantel-Haesnzel Chi-square statistic in the SAS output
  - `CMH`: "*Row Mean Scores Differ*" statistic.
  - The CMH values would now display different values unlike the 2×2 tables.

2.8 Example

The data shown in the table comes from a randomized, double-blind clinical trial investigating a new treatment for rheumatoid arthritis. Investigators compared the new treatment with a placebo; the response measured was whether there was no, some, or marked improvement in the symptoms of rheumatoid arthritis when the active treatment was administered.

- Is there sufficient evidence to conclude that the active treatment group has higher improvement compared to the placebo group?

Table 4.1 Combined Rheumatoid Arthritis Data

	Improvement			
Treatment	None	Some	Marked	Total
Active	13	7	21	41
Placebo	29	7	7	43
Total	42	14	28	84

2.9 Example

The data shown in the table comes from a randomized, double-blind clinical trial investigating a new treatment for rheumatoid arthritis. Investigators compared the new treatment with a placebo; the response measured was whether there was no, some, or marked improvement in the symptoms of rheumatoid arthritis when the active treatment was administered.

- Is there sufficient evidence to conclude that the active treatment group has higher improvement compared to the placebo group?

2.9.1 SAS Code

```
data arth;
input treat $ response $ count @@;
datalines;
active none 13 active some 7 active marked 21
placebo none 29 placebo some 7 placebo marked 7
;
proc freq data=arth order=data;
weight count;
tables treat*response / cmh chisq nocol nopct;
run;
```

2.9.2 R Code

```
library(vcdExtra)
dat  <- expand.grid(Treat=c("Active","Placebo"),
                    Response = c("None","Some","Marked")
                  ) |>
mutate(count = c(13,29,7,7,21,7))

xtabs(count ~  Treat+Response  ,data=dat) -> dat_cc
dat_cc
```

```
         Response
Treat      None Some Marked
  Active     13    7     21
  Placebo    29    7      7
```

```r
CMHtest(dat_cc,
        types="ALL",
        overall=T)
```

```
Cochran-Mantel-Haenszel Statistics for Treat by Response

                 AltHypothesis  Chisq Df       Prob
general    General association 12.900  2 0.00158084
rmeans  Row mean scores differ 12.859  1 0.00033586
cmeans  Col mean scores differ 12.900  2 0.00158084
cor        Nonzero correlation 12.859  1 0.00033586
```

2.9.3 Answer

Q_S = 12.86, p=0.0003. We have strong evidence to conclude that the active treatment displayed higher improvement compared to the placebo group.

- The active group has a higher proportion of marked + some improvement, which enables us to formulate the following conclusion.

2.10 Exercise

A survey was administered to students at a US-based university who are habitual smokers to determine the likelihood of using a mobile app to help them quit smoking. Is there sufficient evidence to conclude that heavy smokers are less likely to use a mobile app compared to light smokers?

```
       Response
Smokers Very Unlikely Unlikely Likely Very Likely
  Heavy            36       75     56          17
  Light            38       59     64          23
```

2.11 Exercise

A survey was administered to students at a US-based university who are habitual smokers to determine the likelihood of using a mobile app to help them quit smoking. Is there sufficient evidence to conclude that heavy smokers are less likely to use a mobile app compared to light smokers?

	Very Unlikely	Unlikely	Likely	Very Likely
Heavy	36	75	56	17
Light	38	59	64	23

2.11.1 SAS Code

```
data smoke;
input treat $ response $ count @@;
datalines;
heavy vu 36 heavy u 75 heavy l 56 heavy vl 17
light vu 38 light u 59 light l 64 light vl 23
;

proc freq data=smoke order=data;
weight count;
tables treat*response / cmh chisq nocol nopct;
run;
```

2.11.2 R Code

```
library(vcdExtra)
CMHtest(dat_cc,
        types="rmeans",
        overall=T)
```

```
Cochran-Mantel-Haenszel Statistics for Smokers by Response

                 AltHypothesis  Chisq Df    Prob
rmeans Row mean scores differ 1.0394  1 0.30795
```

2.11.3 Answer

$Q_S = 1.04$, p=0.31. There is insufficient evidence that heavy smokers are less likely to use a mobile app compared to light smokers.

3 Sets of $2 \times r$ tables

3.1 Sets of $2 \times r$ tables

- We extend the Mantel-Haenszel strategy to $2 \times r$ tables for ordinal responses
- Compute mean scores for the responses
  - Calculate mean score differences across tables
  - Similar approach to the difference in proportions (Q_S)
- Assume hypergeometric distributions for the contingency tables in the different strata.
  - Also assume that the across-strata sample sizes are sufficiently large for each group (row sums)

3.2 Extended Mean Score Statistic

- Q_{SMH} is the extended Mantel-Haenszel mean score statistic.
  - Also known as the ANOVA statistic
  - Function of the weighted average of the differences in mean scores of the two treatments for all strata.
- Q_{SMH} is good for detecting consistent patterns of differences across strata.
- Sample size requirements involve collapsing each stratum table into a 2x2 table by choosing cutpoints (Just like that in Q_S), then apply the Mantel-Fleiss criterion.

3.3 Different Types of Scores

3.3.1 Integer Scores

- (0,1,2,...)
- Response levels are ordered categories that correspond to discrete counts.
- Produce same results regardless of starting number.
- Default in SAS.

3.3.2 Rank

- Rank scores based on the row totals.
- In stratified analysis, rank scores are not robust when there are large strata.

$$R_i^1 = \sum_{k<i} n_{k+} + \frac{n_{i+} + 1}{2}$$

Ridit - Ridit scores are standardized rank scores based on the sample size

$$R_i^3 = 1/(n+1) \left[\sum_{k<i} n_{k+} + \frac{n_{i+} + 1}{2} \right]$$

3.3.3 Standardized Midranks

- Also known as modified ridit scores.
- Constrained to lie between 0 and 1.
- Requires no scaling of response levels.
- Used when there is no argument for equal spaces between ordinal levels

$$R_i^3 = 1/(n+1) \left[\sum_{k<i} n_{k+} + \frac{n_{i+} + 1}{2} \right]$$

3.3.4 Logrank Scores

- Useful when the distribution is L-shaped.
- Need to apply the logarithmic transformation of the rank scores before applying to the `tables` statement.

3.4 Different Types of Scores

> **Note**
>
> When there is no strata, the rank, ridit, and modridit scores correspond to the Wilcoxon rank-sum test.

> **Note**
>
> In stratified analyses, the *modridit scores* are preferred out of these rank-based scores because it produces an extension of the Wilcoxon rank-sum test.

3.5 Measures of Association in Sets of $ 2 $\times$r$ tables

3.5.1 Somers' D

- Measure of association for ordinally scaled data.
- For ordinal scores with two groups, let $G1$ be the score of group 1 and $G2$ be the score of group 2

$$D = \frac{P(G1 > G2) - P(G1 < G2)}{P(G1 > G2) + P(G1 < G2) + P(G1 = G2)}$$

> **Note**
>
> SAS will provide two values of the Somers'D: $D(R|C)$ and $D(C|R)$
>
> - $D(R|C)$ treats the *column* as the independent variable.
> - $D(C|R)$ treats the *row* as the independent variable

3.5.2 Mann-Whitney rank measure of association

- Assesses association between a dichotomous predictor variable and an ordinal outcome.
- Can be calculated from Somers'D

$$g_h = (D + 1)/2$$

where the standard error of g_h is $v_h = v_D/2$.

3.6 Extension of the Mann-Whitney Rank Measure

- When there are q strata, we can calculate a test statistic Q_H such that

$$Q_H = \sum_{h=1}^{q} (g_h - \tilde{g})^2 / v_h$$

$$\tilde{g} = \frac{\sum_{h=1}^{q} g_h \left(n_{h1} n_{h2} / (n_{h1} + n_{h2})\right)}{\sum_{h=1}^{q} \left(n_{h1} n_{h2} / (n_{h1} + n_{h2})\right)}$$

- This is an extension of the van Elteren's rank sum test.

3.7 Scores: SAS

- The default score used in SAS are integer/table scores
- Scoring method can be specified using the SCORES option in SAS
  - TABLE: integer
  - RIDIT: midranks
  - MODRIDIT: standardized midranks
  - RANK: rank scores.
- Use the Cochran-Mantel-Haenszel table (CMH) in testing the hypothesis
  - Difference in mean scores: "Row means differ"

```
proc freq;
tables A*B/scores=MODRIDIT;
run;
```

3.8 Somers' D: SAS

- `measures` includes the Somers' D estimate with the standard errors (`ASE` column).
- The Mann-Whitney rank-sum statistic can be calculated using PROC IML.
- The `cl` option calculates confidence intervals for Somers' D

```
proc freq;
tables A*B/scores=MODRIDIT measures cl;
run;
```

3.9 Mann-Whitney: SAS (optional)

```
proc freq ;
weight count;
tables S*A*B/ measures scores=modridit;
output out=somerout smdcr;                                    (1)
ods output CrosstabFreqs=myFreqs;                             (2)
run;
data ns;
set myFreqs; where _type_="110";
run;

proc iml;
use somerout var{_SMDCR_ E_SMDCR};
read all into combined;
use ns var{Frequency};
read all into total;
print total;
WH=(total[1,]#total[2,]) / (total[1,]+total[2,]) //
(total[3,]#total[4,]) / (total[3,]+total[4,]);
print wh;
GH=(combined[,1]+1)/2;
SEGH=(combined[,2])/2;
print wh GH SEGH;
QH= (GH[1] - GH[2])**2 / SSQ(SEGH);
pvalue=1-probchi(QH,1);
print GH SEGH QH pvalue;
gtilde1=sum(GH # WH);
gtilde2=sum(WH);
gtilde=gtilde1/gtilde2;
gtildevar=(sum(WH##2 # SEGH##2)) /((sum(WH))**2);
segtilde=sqrt(gtildevar);
qgtilde=(gtilde-.5)**2/gtildevar;
print gtilde gtildevar qgtilde;
quit;
```

(1) Outputs somers'D(C|R) and error
(2) crosstabs outputs the contigency table

3.10 Example

The data shown in the table comes from a randomized, double-blind clinical trial investigating a new treatment for rheumatoid arthritis. Investigators compared the new treatment with a placebo; the response measured was whether there was no, some, or marked improvement in the symptoms of rheumatoid arthritis when the active treatment was administered.

- Is there sufficient evidence to conclude that the active treatment group has higher improvement compared to the placebo group after controlling for gender?
- Compare the results using table scores and modridit scores.

Table 4.2 Rheumatoid Arthritis Data

		Improvement			
Gender	Treatment	None	Some	Marked	Total
Female	Active	6	5	16	27
Female	Placebo	19	7	6	32
Total		25	12	22	59
Male	Active	7	2	5	14
Male	Placebo	10	0	1	11
Total		17	2	6	25

3.11 Example

- Is there sufficient evidence to conclude that the active treatment group has higher improvement compared to the placebo group after controlling for gender?
- Compare the results using table scores and modridit scores.

Table 4.2 Rheumatoid Arthritis Data

Gender	Treatment	Improvement None	Some	Marked	Total
Female	Active	6	5	16	27
Female	Placebo	19	7	6	32
Total		25	12	22	59
Male	Active	7	2	5	14
Male	Placebo	10	0	1	11
Total		17	2	6	25

3.11.1 SAS Code

```
data arth;
input gender $ treat $ response $ count @@;
datalines;
female active none 6 female active some 5 female active marked 16
female placebo none 19 female placebo some 7 female placebo marked 6
male active none 7 male active some 2 male active marked 5
male placebo none 10 male placebo some 0 male placebo marked 1
;

proc freq data=arth order=data;
weight count;
tables gender*treat*response / cmh nocol nopct;                    ①
tables gender*treat*response / cmh nocol nopct scores=modridit;    ②
run;
```

① Table Scores
② Modridit scores (Standardized Midranks)

3.11.2 R Code

```
dat <- expand.grid(Improvement=c("None","Some","Marked"),
            Treatment = c("Active","Placebo"),
```

```r
           gender=c("Female","Male")
           ) |>
mutate(count = c(6,5,16,19,7,6,7,2,5,10,0,1))
xtabs(count ~  Treatment + Improvement + gender  ,data=dat) -> dat_cc
dat_cc
CMHtest(dat_cc,
        types="ALL",
        cscores = 1:ncol(dat_cc),                                    # (1)
        overall=T, strata="gender")

CMHtest(dat_cc,
        types="ALL",
        overall=T,
        cscores="midrank",                                           # (2)
        strata="gender")
```

1. table scores
2. standardized midrank scores

```
, , gender = Female

         Improvement
Treatment None Some Marked
  Active     6    5     16
  Placebo   19    7      6

, , gender = Male

         Improvement
Treatment None Some Marked
  Active     7    2      5
  Placebo   10    0      1

$`gender:Female`
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
    in stratum gender:Female

                 AltHypothesis  Chisq Df       Prob
general    General association 11.105  2 0.00387828
rmeans  Row mean scores differ 10.935  1 0.00094361
cmeans  Col mean scores differ 11.105  2 0.00387828
cor        Nonzero correlation 10.935  1 0.00094361
```

```
$`gender:Male`
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
    in stratum gender:Male

                 AltHypothesis  Chisq Df     Prob
general    General association 4.7105  2 0.094871
rmeans  Row mean scores differ 3.7128  1 0.053997
cmeans  Col mean scores differ 4.7105  2 0.094871
cor        Nonzero correlation 3.7128  1 0.053997


$ALL
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
    Overall tests, controlling for all strata

                 AltHypothesis  Chisq Df       Prob
general    General association 14.632  2 0.00066473
rmeans  Row mean scores differ 14.632  1 0.00013068
cmeans  Col mean scores differ 14.632  2 0.00066473
cor        Nonzero correlation 14.632  1 0.00013068


$`gender:Female`
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
    in stratum gender:Female

                 AltHypothesis  Chisq Df       Prob
general    General association 11.105  2 0.00387828
rmeans  Row mean scores differ 10.879  1 0.00097249
cmeans  Col mean scores differ 11.105  2 0.00387828
cor        Nonzero correlation 10.879  1 0.00097249


$`gender:Male`
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
    in stratum gender:Male

                 AltHypothesis  Chisq Df     Prob
general    General association 4.7105  2 0.094871
rmeans  Row mean scores differ 4.1469  1 0.041712
cmeans  Col mean scores differ 4.7105  2 0.094871
```

```
cor        Nonzero correlation 4.1469  1 0.041712


$ALL
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
    Overall tests, controlling for all strata

                 AltHypothesis  Chisq Df       Prob
general    General association 14.632  2 0.00066473
rmeans  Row mean scores differ 15.004  1 0.00010728
cmeans  Col mean scores differ 14.632  2 0.00066473
cor        Nonzero correlation 15.004  1 0.00010728
```

3.11.3 Answer

$Q_{SMH} = 14.6$, p<0.001. There is sufficient evidence to conclude that the active treatment group has higher improvement compared to the placebo group.

When using the modridit scores, $Q_{SMH} = 15.00, p < 0.001$. Same conclusion with a slight deviation between statistics.

3.12 Exercise

The following data is obtained from the 2003 Programme for International Student Assessment (PISA) study.

- Use the appropriate analysis to determine whether the attitude toward school is different between students in the US and Korea after controlling for gender.
- Calculate the relevant Somers' D values for each gender.

3.12.1 Contingency Table

		To what extent do you agree that school has been a waste of time?			
Country	*Gender*	*Strongly agree*	*Agree*	*Disagree*	*Strongly disagree*
United States	Females	38	125	1227	1253
	Males	102	249	1369	922
Korea	Females	26	199	1415	578
	Males	58	281	1942	897

3.12.2 SAS Data

```
data school;
input gender $ location $ response $ count @@;
datalines;
female US sd 1253 female US d 1227 female US a 125 female US sa 38
female korea sd 578 female korea d 1415 female korea a 199 female korea sa 26
male US sd 922 male US d 1369 male US a 249 male US sa 102
male korea sd 897 male korea d 1942 male korea a 281 male korea sa 58
;
```

3.13 Exercise

The following data is obtained from the 2003 Programme for International Student Assessment (PISA) study.

- Use the appropriate analysis to determine whether the attitude toward school is different between students in the US and Korea after controlling for gender.
- Calculate the relevant Somers' D values for each gender.

3.13.1 SAS Code

```
proc freq data=school order=data;
weight count;
tables gender*location*response/all nocol nopct measures cl;
tables gender*location*response/all nocol nopct measures scores=modridit cl;
run;
```

3.13.2 R Code

```
dat <- expand.grid(Rating=c("StD","D","A","StA"),
            gender=c("Female","Male"),
            country = c("US","Korea")) |>
mutate(count = c(1253,1227,125,38,922,1369,249,102,578,1415,199,26,897,1942,281,58))

xtabs(count ~  country +Rating + gender  ,data=dat) -> dat_cc
dat_cc
```

```
, , gender = Female

       Rating
country  StD    D    A  StA
  US     1253 1227  125   38
  Korea   578 1415  199   26

, , gender = Male

       Rating
country  StD    D    A  StA
  US      922 1369  249  102
  Korea   897 1942  281   58
```

```
CMHtest(dat_cc,
        types="ALL",
        cscores = 1:ncol(dat_cc),
        overall=T, strata="gender")
```

```
$`gender:Female`
Cochran-Mantel-Haenszel Statistics for country by Rating
    in stratum gender:Female

                 AltHypothesis  Chisq Df       Prob
general    General association 246.04  3 4.6999e-53
rmeans  Row mean scores differ 182.24  1 1.5754e-41
cmeans  Col mean scores differ 246.04  3 4.6999e-53
cor        Nonzero correlation 182.24  1 1.5754e-41


$`gender:Male`
Cochran-Mantel-Haenszel Statistics for country by Rating
    in stratum gender:Male

                 AltHypothesis   Chisq Df       Prob
general    General association 64.7130  3 5.7775e-14
rmeans  Row mean scores differ  1.2096  1 2.7140e-01
cmeans  Col mean scores differ 64.7130  3 5.7775e-14
cor        Nonzero correlation  1.2096  1 2.7140e-01


$ALL
```

```
Cochran-Mantel-Haenszel Statistics for country by Rating
    Overall tests, controlling for all strata

                 AltHypothesis  Chisq Df       Prob
general    General association 246.36  3 4.0051e-53
rmeans  Row mean scores differ 91.317  1  1.224e-21
cmeans  Col mean scores differ 246.36  3 4.0051e-53
cor        Nonzero correlation 91.317  1  1.224e-21
```

```
CMHtest(dat_cc,
        types="ALL",
        overall=T,
        cscores="midrank",
        strata="gender")
```

```
$`gender:Female`
Cochran-Mantel-Haenszel Statistics for country by Rating
    in stratum gender:Female

                 AltHypothesis  Chisq Df       Prob
general    General association 246.04  3 4.6999e-53
rmeans  Row mean scores differ 223.84  1 1.3178e-50
cmeans  Col mean scores differ 246.04  3 4.6999e-53
cor        Nonzero correlation 223.84  1 1.3178e-50


$`gender:Male`
Cochran-Mantel-Haenszel Statistics for country by Rating
    in stratum gender:Male

                 AltHypothesis   Chisq Df       Prob
general    General association 64.7130  3 5.7775e-14
rmeans  Row mean scores differ  8.4411  1 3.6683e-03
cmeans  Col mean scores differ 64.7130  3 5.7775e-14
cor        Nonzero correlation  8.4411  1 3.6683e-03


$ALL
Cochran-Mantel-Haenszel Statistics for country by Rating
    Overall tests, controlling for all strata

                 AltHypothesis  Chisq Df       Prob
```

```
general    General association 246.36  3 4.0051e-53
rmeans  Row mean scores differ 149.89  1 1.8367e-34
cmeans  Col mean scores differ 246.36  3 4.0051e-53
cor        Nonzero correlation 149.89  1 1.8367e-34
```

3.13.3 Answer

Using MODRIDIT scores, we get $Q_{SMH} = 149.88, p < 0.001$. Using TABLE scores, we get $Q_{SMH} = 91.3, p < 0.001$. There is sufficient evidence that there is a difference in the attitude toward school between the students from US and Korea.

- I would use the modridit scores because the spacing between ratings is non-uniform.
- You can use multiple scores and see whether the resulting inference is the same.
- The Somers' D (C|R) for females is 0.22 (0.19,0.25)
- The Somer's D (C|R) for males is 0.04 (0.01,0.07)

4 $s \times 2$ table

4.1 $s \times 2$ table

- Used when there are multiple groups defined by an ordinal variable with a dichotomous outcome.
- Example: Degree of gambling experience, degree of smoking experience, Educational Attainment

4.2 $s \times 2$ table

Assume X is an ordinal variable.

To account for ordinality, we assign a set of scores **c** reflecting the relationship between response levels.

$$\mathbf{c} = \{c_1, c_2, c_3, ...c_s\} : c_1 < c_2 < ... < c_s$$

In $2 \times r$ tables, $\mathbf{a} = (a_1, a_2)$ is effectively (0,1)

	A_1	A_2	Total
X_1	n_{11}	n_{12}	n_{1+}
X_2	n_{21}	n_{22}	n_{2+}
...	...	...	...
X_s	n_{s1}	n_{s2}	n_{s+}
Total	n_{+1}	n_{+2}	n

4.3 $s \times 2$ table

The expected frequency can be expressed as

$$\bar{f} = \sum_{i=1}^{s} \sum_{j=1}^{2} \frac{c_i a_j n_{ij}}{n}$$

4.4 $s \times 2$ table

The null hypothesis H_0: There is no association between the two variables. To test this hypothesis, we can use a randomization statistic Q_{CS} such that

$$Q_{CS} = \frac{(\bar{f} - \mu_c \mu_a)^2}{v_c v_a/(n-1)} = (n-1) r_{ac}^2$$

> **Note**
>
> - μ_c is the expected value of the row scores
> - μ_a is the expected value of the column scores.
> - v_c is the variance of the row scores
> - v_a is variance of the column scores

ℹ Note

Q_{CS} is also referred to the correlation statistic. Q_{CS} follows a chi-square distribution with one degree of freedom.

ℹ Note

Q_{CS} is related to the Pearson correlation coefficient r_{ac}

4.5 Cochran-Armitage test

A similar statistic to the correlation statistic is the z^2 statistic used in the Cochran-Armitage test.

- The Cochran-Armitage trend test tests for trends in binomial proportions across levels of an ordinal covariate.

$$z^2 = Q_{CS}\left[\frac{n}{n-1}\right]$$

4.6 $s \times 2$ analysis: SAS

- The correlation statistic can be found in the SAS Output as the Mantel-Haenszel chi-square statistic from the `chisq` option.
- SAS can also perform the Cochran-Armitage trend test by including the `trend` option in the `tables` statement.
- The Pearson correlation coefficient and the Somers' D(C|R) are of particular interest in the `measures` output since both metrics account for ordinality.

```
proc freq;
weight count;
tables a*b/trend chisq measures;
run;
```

4.7 Example

A random sample of 200 married men, all retired, were classified according to education and number of children. The data is shown in the table.

- Is there sufficient evidence of association between the father's educational attainment and the number of children?
- Using the Cochran-Armitage trend test to determine if there is an increasing trend in number of children with increasing educational attainment.

```
dat <- expand.grid(education=c("secondary","college","graduate"), Kids = c("At most 3","More
mutate(count=c(51,61,29,32,17,10))

xtabs(count ~  education+Kids  ,data=dat) -> dat_cc
dat_cc
```

```
           Kids
education   At most 3 More than 3
  secondary        51          32
  college          61          17
  graduate         29          10
```

4.8 Example

A random sample of 200 married men, all retired, were classified according to education and number of children. The data is shown in the table.

- Is there sufficient evidence of association between the father's educational attainment and the number of children?
- Using the Cochran-Armitage trend test to determine if there is an increasing trend in number of children with increasing educational attainment.

4.8.1 SAS Code

```
data educ;
input education $ kids3 $ count @@;
datalines;
secondary leq 51 secondary g 32
college leq 61 college g 17
graduate leq 29 graduate g 10
;
run;

proc freq data= educ order=data;
weight count;
```

```
tables education*kids3/chisq measures trend cl;   /* (1) */
run;
```

(1) `chisq` gives Q_{CS}, `measures` provides Somers and pearson, `trend` gives cochran-armitage, `cl` calculates confidence intervals.

4.8.2 Answer

$Q_{CS} = 3.47, p = 0.06$. There is weak evidence of association between the father's educational attainment and number of offspring.

The one-sided Cochran-Armitage test yielded a p-value = 0.03, which is sufficient evidence to support a decreasing trend in number of children with increasing educational attainment.

5 Sets of $s \times 2$ tables

5.1 Sets of $s \times 2$ tables

- Just like 2xr tables, there is also a statistic for the association of two ordinal variables for a combined set of strata called the extended Mantel-Haenszel correlation statistic (Q_{CSMH}).
  - Q_{CSMH} approximately follows a chi-square distribution with df=1 when the combined strata sample sizes is greater than or equal to 40 (sufficiently large).

5.2 Example

The table contains two contingency tables of risk perception (minimal, moderate, and substantial) and adolescent tobacco usage by father's usage.

- Comment on whether there is an association between risk perception and adolescent usage when the father does not use tobacco.
- Is there a discernible trend in the proportions of adolescent usage over the levels of risk perception after controlling for father's tobacco usage?

Table 4.5 Adolescent Smokeless Tobacco Usage

Father's Usage	Risk Perception	Adolescent Usage		Total
		No	Yes	
No	Minimal	59	25	84
No	Moderate	169	29	198
No	Substantial	196	9	205
Yes	Minimal	11	8	19
Yes	Moderate	33	11	44
Yes	Substantial	22	2	24

5.3 Example

The table contains two contingency tables of risk perception (minimal, moderate, and substantial) and adolescent tobacco usage by father's usage.

- Comment on whether there is an association between risk perception and adolescent usage when the father does not use tobacco.
- Is there a discernible trend in the proportions of adolescent usage over the levels of risk perception after controlling for father's tobacco usage?

5.3.1 Contingency table

Table 4.5 Adolescent Smokeless Tobacco Usage

Father's Usage	Risk Perception	Adolescent Usage		Total
		No	Yes	
No	Minimal	59	25	84
No	Moderate	169	29	198
No	Substantial	196	9	205
Yes	Minimal	11	8	19
Yes	Moderate	33	11	44
Yes	Substantial	22	2	24

5.3.2 SAS Code

```
data tobacco;
length risk $11. ;
input f_usage $ risk $ usage $ count @@;
datalines;
no minimal no 59 no minimal yes 25
no moderate no 169 no moderate yes 29
no substantial no 196 no substantial yes 9
yes minimal no 11 yes minimal yes 8
yes moderate no 33 yes moderate yes 11
yes substantial no 22 yes substantial yes 2
;
proc freq;
weight count;
tables f_usage*risk*usage /cmh chisq measures trend;
tables f_usage*risk*usage /cmh scores=modridit;
run;
```

5.3.3 Answer

When the father is not a tobacco user, $Q_{CS} = 34.3, p < 0.001$, which means there is sufficient evidence of an association between risk perception and participant tobacco use.

$Q_{CSMH} = 39.3$ using modified ridit scores ($Q_{CSMH} = 40.7$ for table scores), resulting to a p-value p<0.001. There is sufficient evidence of an association between risk perception and participant tobacco use.

6 $s \times 2$: Exact Analysis

6.1 Exact Analysis

- For small sample sizes, Q_{CS} does not satisfy the asymptotic requirements.
- We use exact tests to calculate the correct Mantel-Haenszel statistics.
- We manually calculate the Wilcoxon rank and logrank scores for each row.
  - Wilcoxon vs logrank: Logrank is better when there are more counts in the lower end of the row ordinal variable.

6.2 Exact Analysis: SAS

- The `exact` statement is used to perform an exact test on the correlation statistic.
  - `exact mhchi` would perform the statement on the Mantel-Haenszel statistic.
  - `scorout` outputs the scores used in the exact analysis

```
proc freq;
tables a*b /scorout;
exact mhchi;
run;
```

6.3 Example

The table shows data from a study of mice who were exposed to a bacterial challenge and treated with the drugs carbenicillin or cefotaxmine (Bowdre, et al. 1983). Surviving mice were assessed at intervals of 6 to 24 hours. Is there evidence that there is a difference in performance between the two drugs?

6.3.1 Contingency Table

Table 4.9 Mice Surviving Exposure to Vibrio Vulnificus

Hours	Carbenicillin	Cefotaxime	Total	Ranks	Logranks
0–6	1	1	2	1.5	0.909
6–12	3	1	4	4.5	0.709
12–18	5	1	6	9.5	0.334
18–24	1	0	1	13	0.234
24–30	1	2	3	15	−0.099
30–48	0	2	2	17.5	−0.433
48–72	1	1	2	19.5	−0.933
72–96	0	1	1	21	−1.433
> 96	0	1	1	22	−2.433
Total	12	10	22		

6.3.2 SAS Data

```
data mice;
input LogRank Treatment $ count @@;
datalines;
0.909 Ca 1 0.909 Ce 1
0.709 Ca 3 0.709 Ce 1
0.334 Ca 5 0.334 Ce 1
0.234 Ca 1 0.234 Ce 0
-0.099 Ca 1 -0.099 Ce 2
-0.433 Ca 0 -0.433 Ce 2
-0.933 Ca 1 -0.933 Ce 1
-1.433 Ca 0 -1.433 Ce 1
-2.433 Ca 0 -2.433 Ce 1
;
```

6.4 Example

The table shows data from a study of mice who were exposed to a bacterial challenge and treated with the drugs carbenicillin or cefotaxmine (Bowdre, et al. 1983). Surviving mice were assessed at intervals of 6 to 24 hours. Is there evidence that there is a difference in performance between the two drugs?

6.4.1 SAS Code


```
data mice;
input LogRank Treatment $ count @@;
datalines;
0.909 Ca 1 0.909 Ce 1
0.709 Ca 3 0.709 Ce 1
0.334 Ca 5 0.334 Ce 1
0.234 Ca 1 0.234 Ce 0
-0.099 Ca 1 -0.099 Ce 2
-0.433 Ca 0 -0.433 Ce 2
-0.933 Ca 1 -0.933 Ce 1
-1.433 Ca 0 -1.433 Ce 1
-2.433 Ca 0 -2.433 Ce 1
;
```

```
proc freq order=data;
weight count;
tables LogRank*Treatment / norow nocol nopct scorout chisq;        (1)
tables LogRank*Treatment / noprint scores=rank scorout chisq;      (2)
exact mhchi;
run;
```

(1) Using Logranks
(2) using ranks. Note that logranks preserves the order of the original data set.

6.4.2 Answer

The exact logrank analysis showed that $Q_{CS} = 4.06, p = 0.04$, which means we have sufficient evidence of an association/difference between the treatment and specimen survival.

However, the exact *rank* analysis showed that $Q_{CS} = 3.51, p = 0.06$, which means we have weak evidence of an association/difference between the treatment and specimen survival.

I would favor the logrank analysis better because it encapsulates the data better and accounts for the higher representation of the lower end of the ordinal variable.